ZMYM3 as a Context-Specific Dependency in Atypical Teratoid/Rhabdoid Tumor: Discovery via CRISPR DepMap Screening and Single-Nucleus RNA Sequencing Analysis

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Code available at: https://github.com/JacquelineJialinLi/zmym3-atrt-project

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INTRODUCTION

Atypical teratoid/rhabdoid tumor (AT/RT) is one of the most lethal pediatric brain tumors, predominantly affecting children under three years of age with a five-year overall survival of only 20-50% despite aggressive multimodal therapy.[1,2] Current regimens combining surgical resection, intensive chemotherapy, and craniospinal irradiation carry severe neurodevelopmental toxicity in infants, and no molecularly targeted therapies have been approved.[2,3] This creates an urgent clinical need, and a central biological question: what makes AT/RT cells uniquely vulnerable, and can those vulnerabilities be exploited therapeutically without harming the developing brain?

To systematically identify which genes AT/RT cells are specifically dependent on for survival, we leveraged the Cancer Dependency Map (DepMap), a resource from the Broad Institute that profiles gene essentiality across over 1,200 cancer cell lines using genome-scale CRISPR-Cas9 loss-of-function screens.[4-7] In these screens, each gene is individually knocked out and its effect on cell fitness quantified. AT/RT is uniquely suited to this approach: the disease is defined in over 95% of cases by biallelic inactivation of SMARCB1, a core subunit of the SWI/SNF chromatin remodeling complex, and beyond this single event, AT/RT genomes accumulate remarkably few additional mutations.[8-11] The oncogenic state is therefore sustained not by a cascade of genetic alterations but by the aberrant epigenetic programs that follow SWI/SNF dysfunction — meaning there are no secondary mutations to target, and the residual machinery these cells depend on for survival becomes the most logical therapeutic space to interrogate. By comparing DepMap fitness scores in AT/RT cell lines against all other cancer types, one can identify genes that are selectively essential in AT/RT, distinguishing targets that reflect the specific consequences of SMARCB1 loss from those broadly required across cancer types. This distinction is critical: genes essential across many tumor types are likely also required in normal proliferating tissues, carrying substantial toxicity risk, whereas dependencies concentrated in AT/RT are more likely to expose the epigenetic vulnerabilities specific to this disease. A key practical challenge is the limited number of AT/RT cell lines available in DepMap, which constrains statistical power; we address this through a consensus approach requiring agreement across multiple complementary statistical tests to define high-confidence hits.

Using this framework, we identified ZMYM3 (Zinc Finger MYM-Type Protein 3) as a top-ranked AT/RT-specific fitness gene. ZMYM3 is a chromatin-associated zinc finger protein and component of the CoREST transcriptional repressor complex, which regulates gene silencing through histone deacetylation.[12,13] Its emergence as an AT/RT dependency is biologically plausible: in cells that have lost SWI/SNF-mediated chromatin remodeling, CoREST-mediated repression may become disproportionately important for sustaining proliferation. However, identifying a gene as a fitness dependency in cell lines is only a first step — it does not reveal whether that dependency is uniformly distributed across the heterogeneous cellular landscape of primary tumors. AT/RT comprises three epigenetic subgroups — ATRT-TYR, ATRT-SHH, and ATRT-MYC — each with distinct identities and transcriptional programs,[14,15] and recent single-nucleus RNA sequencing (snRNA-seq) has revealed intra-tumoral heterogeneity including a shared cycling progenitor-like population across subgroups.[16,17] Whether ZMYM3 is expressed broadly or selectively has direct implications for whether it represents a universal or stratified therapeutic target. To address this, we analyzed snRNA-seq data from a published primary AT/RT patient cohort[16] to characterize ZMYM3 expression across tumor subgroups and cell states, and to assess whether its context-specific essentiality is accompanied by tumor-enriched transcriptional activity.

RESULTS

DepMap Identification of ZMYM3 as a Context-Specific AT/RT Dependency

Pan-indication context-specific gene discovery. We first sought to build a unified, comparable dataset of gene essentiality across cancer types. We obtained Chronos-corrected

CRISPR gene-effect scores from the DepMap 26Q1 release, encompassing 1,208 cell lines and 18,531 genes. Chronos correction adjusts raw sgRNA counts for copy-number bias and screen-quality differences, yielding scores where 0 indicates no fitness effect and increasingly negative values indicate essentiality. Cell lines were annotated with cancer type using OncotreeSubtype classifications from the accompanying cell model metadata, yielding 167 unique indications. Because rhabdoid tumors are distributed across multiple Oncotree subtypes — Atypical Teratoid/Rhabdoid Tumor (6 cell lines), Rhabdoid Cancer (5 cell lines), and Extrarenal Rhabdoid Cancer (5 cell lines) — we merged all 16 rhabdoid-associated cell lines into a single “Atypical Teratoid/Rhabdoid Tumor” indication to maximize statistical power. This merge assumes that rhabdoid tumors at different anatomic sites share a sufficiently common SMARCB1-loss-driven dependency program to be analyzed jointly — an assumption consistent with their shared genetic basis but one we revisit when interpreting individual hits.

We then asked, for each indication, which genes are required more strongly in that indication than in all others. Because no single statistical test is uniformly powerful — Fisher’s exact test captures binary dependency enrichment, the t-test captures mean shifts, and the Wilcoxon rank-sum test captures distributional shifts without parametric assumptions — we hypothesized that a consensus across all three tests would yield more robust hits than any one alone. We therefore applied all three in parallel for each of the 71 indications with at least 4 cell lines: Fisher’s exact test with dependency binarized at a gene-effect score cutoff of −0.6, the independent two-sample t-test, and the Wilcoxon rank-sum test (excluded for indications with fewer than 10 cell lines due to insufficient power). P-values for each test were corrected for multiple testing using the Benjamini-Hochberg procedure, and the top 10 genes passing a corrected significance threshold of 0.05 were retained per test. Genes identified as significant by the intersection of all applicable tests were designated high-confidence consensus context-specific fitness genes. Across all 71 indications, this framework identified 130 consensus context-specific genes spanning 40 indications, including well-established dependencies that served as positive controls: BRAF and SOX10 in melanoma, KRAS in pancreatic adenocarcinoma, CTNNB1 in colon adenocarcinoma, BCR and ABL1 in chronic myeloid leukemia, and IRF4 in plasma cell myeloma (Figure S1). The identification of these canonical context-specific dependencies suggests that our framework is well-calibrated for identifying biologically meaningful hits.

Three consensus genes emerge for AT/RT: SMARCD1, ZMYM3, and NABP2. For the merged AT/RT indication (N = 16 cell lines vs. 1,192 background), the consensus approach identified three context-specific fitness genes: SMARCD1, ZMYM3, and NABP2 (Figure 1). SMARCD1, a core SWI/SNF subunit, was the strongest hit (mean gene-effect score in AT/RT: −0.817 vs. −0.208 in all other lines; Fisher $p = 1.69 \times 10^{-8}$; t-test $p = 1.96 \times 10^{-12}$; Wilcoxon $p = 3.7 \times 10^{-5}$), with 87.5% of AT/RT cell lines scoring as dependent (vs. 9.4% background). Its identification is biologically expected, as residual SWI/SNF subunits are known to be required in SMARCB1-deficient contexts.[10] ZMYM3 showed a mean score of −0.626 in AT/RT versus −0.139 in background (Fisher $p = 2.26 \times 10^{-6}$; t-test $p = 1.37 \times 10^{-13}$; Wilcoxon $p = 4.1 \times 10^{-5}$), with 56.2% of AT/RT lines dependent versus only 2.9% background (Figure S2). NABP2, a single-stranded DNA binding protein involved in DNA damage repair, showed comparable effect size (mean −0.635 vs. −0.151; 50.0% dependent in AT/RT vs. 1.7% background). Among the three hits identified, ZMYM3 displayed the lowest off-target dependency in other tissues while maintaining significant AT/RT dependency. Notably, EZH2 and EED — components of the PRC2 complex and established therapeutic targets in rhabdoid tumors[11] — ranked among the top 20 genes by multiple individual tests but did not reach consensus across all three, providing further validation of the framework's specificity.

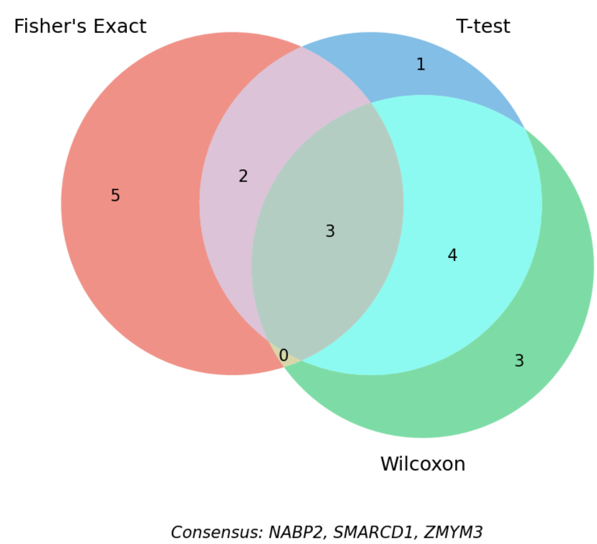

Figure 1. Consensus dependency analysis for AT/RT. Venn diagram showing the overlap of significant hits across the three statistical tests. Three genes reached significance in all three tests: ZMYM3, SMARCD1, and NABP2.

ZMYM3 dependency is specific to rhabdoid tumors. To further characterize ZMYM3 dependency, we examined its gene-effect score distribution across all indications in the DepMap (Figure 2A). The AT/RT merged indication exhibited the most negative median ZMYM3 score among all cancer types. Specifically, AT/RT is significantly more dependent on ZMYM3 (FDR < 0.05) than 68 out of 70 indications, with its mean falling 5.1 standard deviations below the mean of all other indications' per-indication means. Individual AT/RT cell lines ranged from −1.239 (strongly dependent) to −0.069 (non-dependent), indicating heterogeneity in the degree of ZMYM3 dependency across rhabdoid cell lines. Bulk RNA-seq expression data from the CCLE showed that AT/RT cell lines had modestly above-median ZMYM3 transcript expression (median log2[TPM+1] = 4.64 vs. 3.88 globally), suggesting that ZMYM3 is expressed in AT/RT cells but that the magnitude of expression does not drive the observed dependency (Figure 2B).

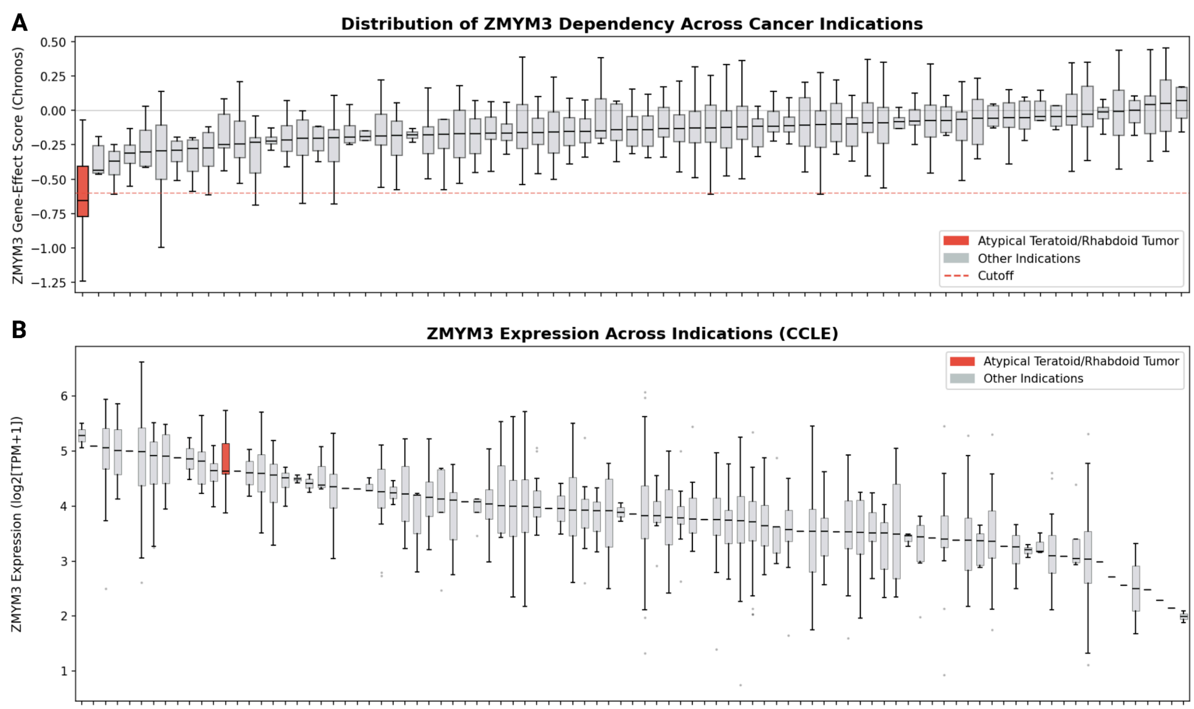

Figure 2. Distribution of ZMYM3 dependency scores and expression across cancer and non-cancer indications. (A) Chronos gene-effect scores for ZMYM3 across both cancer and non-cancer indications in the DepMap. AT/RT cell lines (red) exhibit the most negative scores of any indication, falling well below the −0.6 dependency threshold (dashed red line). **(B)** ZMYM3 expression levels using bulk RNA-seq data from the Cancer Cell Line Encyclopedia (CCLE). AT/RT cell lines (red) exhibit slightly elevated ZMYM3 expression compared to the median and rank highly across all indications.

Co-dependency analysis links ZMYM3 to SWI/SNF components. We computed Pearson correlations between ZMYM3 and all other genes across their dependency profiles in

DepMap cell lines to identify functionally related genes (Figure 3). Among SWI/SNF complex members, SMARCD1 exhibited the strongest positive co-dependency with ZMYM3 ($r = 0.208$, $p = 9.87 \times 10^{-12}$), followed by SMARCC1 ($r = 0.091$, $p = 1.49 \times 10^{-3}$) and ARID1B ($r = 0.073$, $p = 0.011$). Among CoREST complex members, ZMYM4 ($r = 0.094$, $p = 0.001$) and HMG20B ($r = 0.070$, $p = 0.016$) showed modest positive correlations, while KDM1A showed a weak negative correlation ($r = -0.062$, $p = 0.031$). The positive co-dependency between ZMYM3 and multiple SWI/SNF subunits is consistent with the hypothesis that ZMYM3's CoREST-mediated function becomes selectively critical in the context of disrupted SWI/SNF chromatin remodeling.

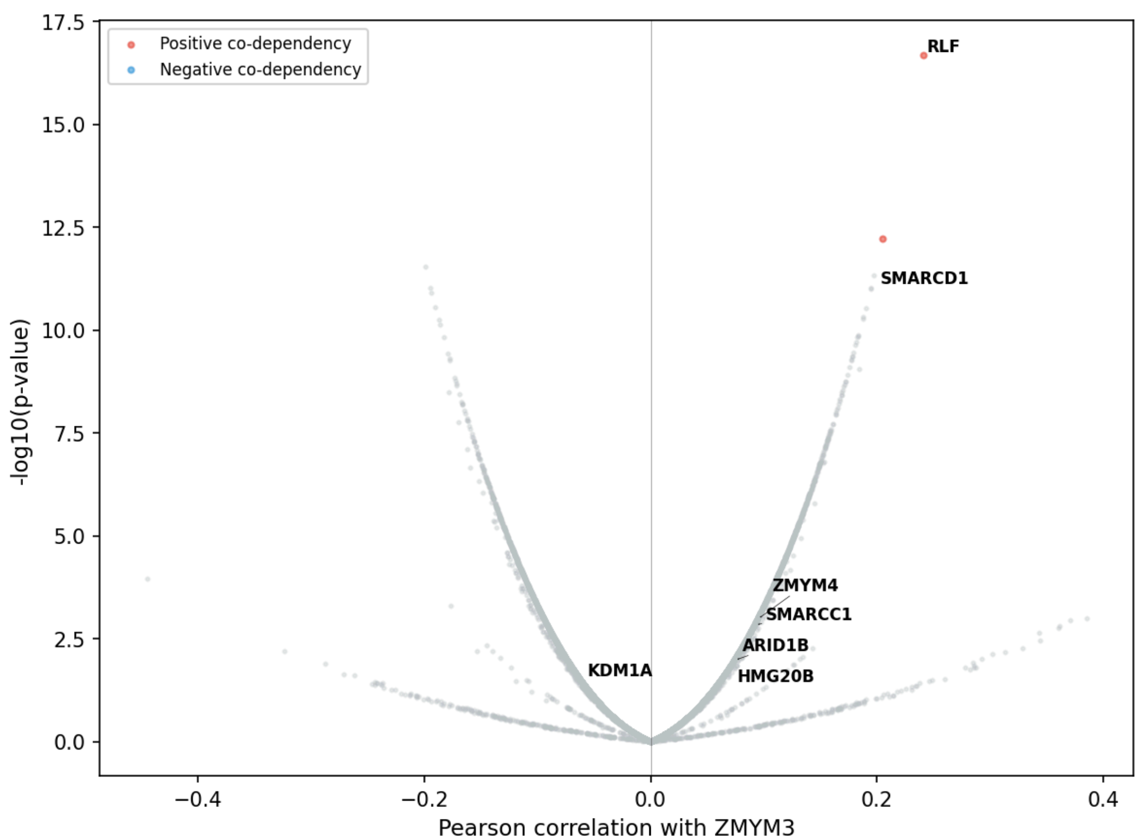

Figure 3. ZMYM3 co-dependency landscape across DepMap cell lines. Volcano plot of Pearson correlation coefficients between ZMYM3 and all other genes (N = 18,434) computed across dependency profiles in DepMap cell lines. Each point represents one gene; the x-axis shows correlation with ZMYM3 and the y-axis shows −log10(p-value). Red points indicate significant positive co-dependencies ($r > 0.2$, $p < 10^{-10}$); blue points indicate significant negative co-dependencies. Labeled genes include members of the SWI/SNF complex (SMARCD1, $r = 0.208$; SMARCC1, $r = 0.091$; ARID1B, $r = 0.073$) and CoREST complex (ZMYM4, $r = 0.094$; HMG20B, $r = 0.070$). The positive co-dependency between ZMYM3 and multiple SWI/SNF subunits supports a functional relationship between CoREST-mediated repression and residual SWI/SNF activity in the SMARCB1-deficient context.

snRNA-seq Characterization of ZMYM3 Expression in Primary AT/RT Tumors

Why examine ZMYM3 expression in primary tumors, and what do we expect to find? The DepMap analysis established that ZMYM3 is a fitness dependency in AT/RT cell lines, but cell lines are an imperfect model of primary tumor biology. A natural follow-up question is whether ZMYM3 is expressed in actual patient tumors — and importantly, whether its expression is enriched in tumor cells relative to the surrounding non-malignant brain tissue. If ZMYM3 were highly and selectively expressed in tumor cells, that would further support its potential as a therapeutic target. Conversely, if it is expressed at comparable levels in normal brain cells, any therapeutic strategy would need to rely on functional dependence rather than differential expression to achieve selectivity. Understanding the cellular distribution of ZMYM3 expression across the heterogeneous AT/RT tumor microenvironment — including its variation across the three epigenetic subgroups — is therefore important for interpreting the DepMap findings and for informing any translational next steps.

Quality control and preprocessing of the AT/RT snRNA-seq atlas. To assess ZMYM3 expression in primary patient tumors, we analyzed the snRNA-seq dataset from Paassen et al. (2025) (GEO: GSE283842), comprising seven AT/RT patient samples processed with the 10x Chromium multiome platform.[16] The deposited dataset had been pre-filtered by the original authors; after applying our QC thresholds (200–6,000 detected genes per nucleus, <20% mitochondrial fraction, Scrublet doublet score ≤0.25), 9,604 high-quality nuclei were retained across all three subgroups (ATRT-SHH: 4 samples; ATRT-MYC: 2 samples; ATRT-TYR: 2 samples). Quality metrics were uniformly high: median genes per nucleus ranged from 1,232–1,820 across samples, and mitochondrial fractions were consistently low (median <0.3%), consistent with high-quality nuclear preparations. Data were normalized to 10,000 counts per nucleus (CP10K) and log1p transformed, and the top 2,000 highly variable genes (HVGs) were selected for dimensionality reduction.

Dimensionality reduction and patient-level batch correction. PCA on the HVG matrix was performed, retaining 30 PCs. Because the seven samples originate from distinct patients, patient-of-origin constitutes a dominant source of technical variability; we therefore applied Harmony batch correction (theta = 2, max iterations = 10) to iteratively adjust PC coordinates and minimize patient-level clustering while preserving biological signal.[18] Post-correction UMAP confirmed successful integration: nuclei from distinct patients interleaved within coherent clusters while AT/RT subgroup separation was maintained, indicating effective batch correction without over-correction of biological distinctions.

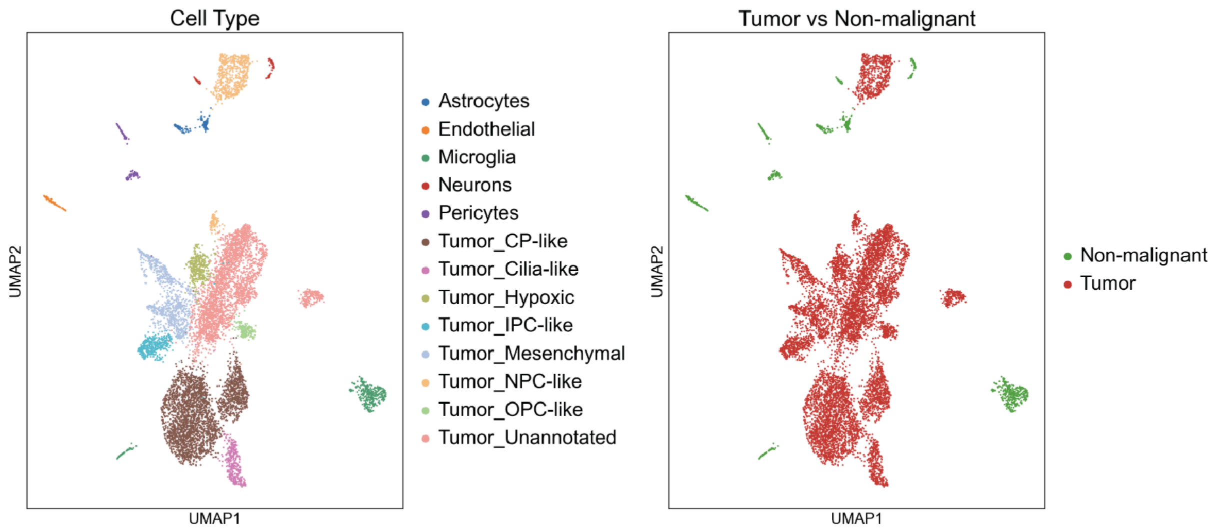

Figure 4. UMAP visualization of 9,604 nuclei colored by cell type annotation (left) and tumor versus non-malignant classification (right). Leiden clustering (resolution = 0.5) on the Harmony-corrected embedding yielded 19 clusters, which were annotated using canonical CNS marker genes and cross-referenced against cell type labels provided by Paassen et al. Tumor populations (8,694 nuclei, 90.5%) encompass multiple differentiation states including choroid plexus-like, neural progenitor cell-like, intermediate progenitor cell-like, cilia-like, mesenchymal-like, OPC-like, and hypoxic clusters. Non-malignant populations (910 nuclei, 9.5%) include microglia, astrocytes, pericytes, endothelial cells, and neurons.

Clustering and cell type annotation. To confirm that our preprocessing and batch correction successfully reproduced the cellular landscape described by Paassen et al., we performed unsupervised clustering and compared our resulting annotations against the original study's deposited metadata. Leiden community detection (resolution = 0.5, n_neighbors = 15) applied to the Harmony-corrected SNN graph yielded 19 clusters (Figure 4), annotated using canonical CNS cell type markers and cross-referenced against the deposited Paassen et al. metadata. The close agreement between our annotations and those of the original authors confirmed successful recapitulation of the published cellular landscape, validating our pipeline as a foundation for downstream ZMYM3 expression analysis. Tumor cells comprised 13 clusters (8,694 nuclei, 90.5%) and were further classified into differentiation states from the original study: CP-like, NPC-like, IPC-like, cilia-like, mesenchymal-like, OPC-like, and hypoxic populations. Non-malignant populations (910 nuclei, 9.5%) included microglia, astrocytes, pericytes, endothelial cells, and neurons — a composition consistent with published snRNA-seq datasets.[19] Median normalized ZMYM3 expression was zero across all cell types, reflecting the high dropout characteristic of lowly expressed chromatin-associated factors in snRNA-seq.

ZMYM3 is expressed at low levels across tumor and non-malignant populations. ZMYM3 expression across the integrated UMAP was sparse and low-level, distributed across both tumor and non-malignant clusters without apparent tumor enrichment.

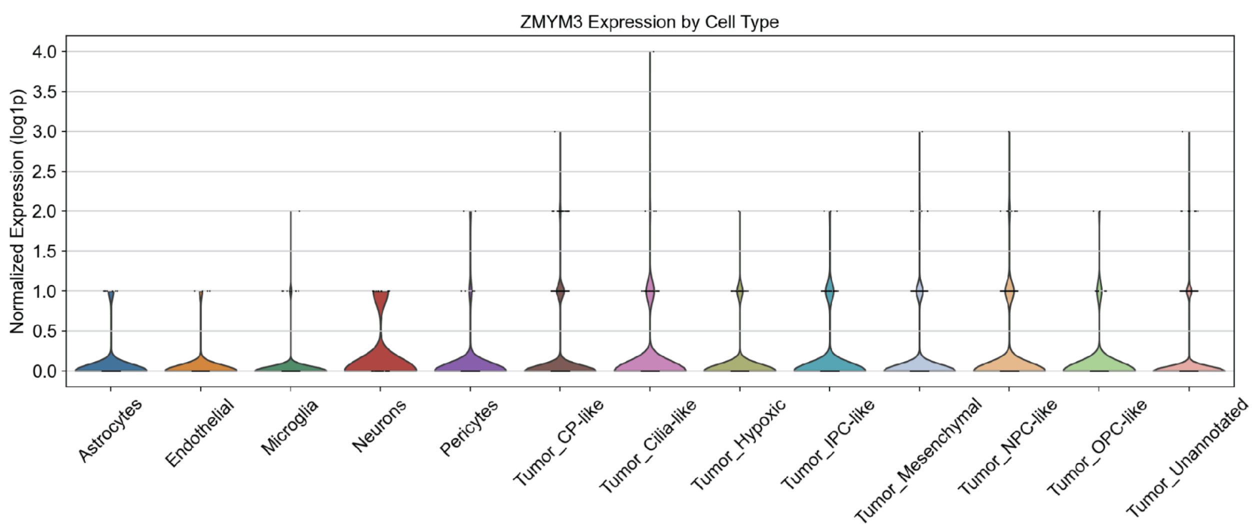

Figure 5. Violin plot of normalized ZMYM3 expression across annotated cell types. Tumor cells (mean = 0.102, 9.3% expressing) do not show enrichment relative to all non-malignant populations; neurons exhibit the highest mean expression (0.172, 17.2% expressing). Wilcoxon rank-sum tests with Benjamini–Hochberg FDR correction were performed comparing tumor cells to each non-malignant cell type (one-sided, tumor > reference). Significant differences (FDR < 0.05) were observed for microglia, endothelial cells, and pericytes, but not for astrocytes (FDR = 0.095) or neurons (FDR = 0.993).

Only 8.9% of all nuclei (9.3% of tumor nuclei) expressed detectable ZMYM3, and quantification per cell type confirmed no tumor-specific enrichment (Figure 5). Wilcoxon rank-sum tests comparing tumor cells to each non-malignant population (BH-FDR corrected) yielded mixed results: tumor expression was modestly higher than in microglia (FDR < 0.001), endothelial cells (FDR = 0.043), and pericytes (FDR = 0.043), but did not differ significantly from astrocytes (FDR

= 0.095) and was lower than in neurons (tumor mean = 0.102 vs. 0.172; FDR = 0.993). The tumor-to-neuron expression ratio of 0.59x further confirms that ZMYM3 lacks preferential tumor expression in this dataset.

ZMYM3 expression shows modest subgroup-level variation. Within the tumor cell fraction, one might ask whether ZMYM3 expression is more pronounced in a particular AT/RT subgroup, which could inform whether its dependency is restricted to a subset of patients or is broadly relevant. Kruskal-Wallis testing revealed a statistically significant difference in ZMYM3 expression across AT/RT subgroups ($H = 10.43$, $p = 5.43 \times 10^{-3}$), with ATRT-TYR tumor cells showing modestly higher expression (mean = 0.116, 10.4% expressing) than ATRT-MYC (mean = 0.088, 8.2%; FDR = 0.025) and ATRT-SHH (mean = 0.093, 8.5%; FDR = 0.016) (Figure 6).

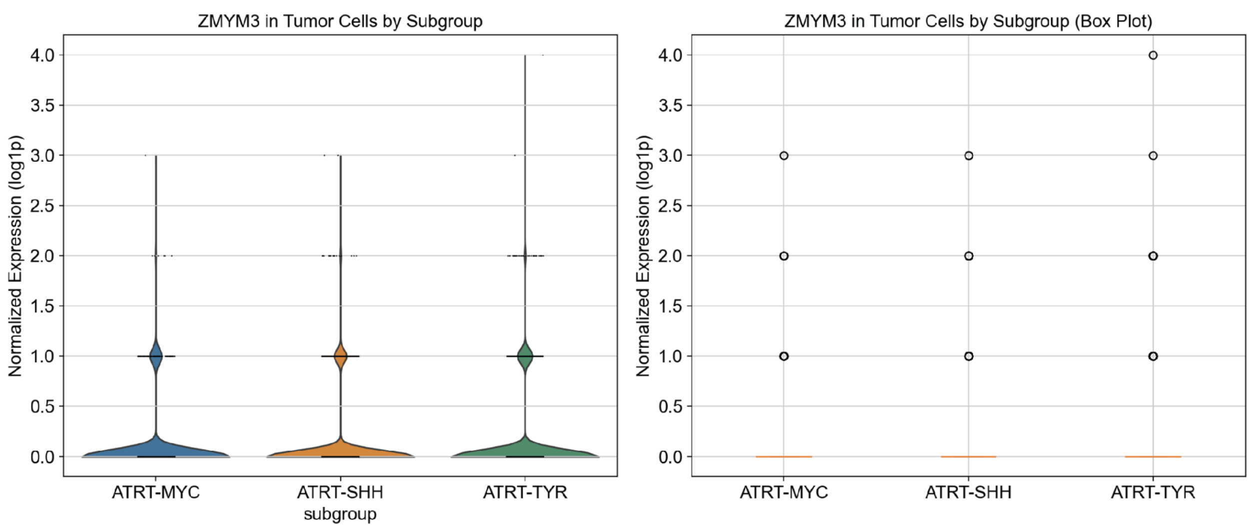

Figure 6. ZMYM3 expression across AT/RT subgroups within tumor cells. ATRT-TYR shows modestly higher expression (mean = 0.116, 10.4% expressing) compared to ATRT-MYC (mean = 0.088, 8.2%) and ATRT-SHH (mean = 0.093, 8.5%). Kruskal–Wallis test: $H = 10.43$, $p = 0.005$. Post-hoc pairwise Wilcoxon tests with BH-FDR correction indicated significant differences between ATRT-TYR and ATRT-MYC (FDR = 0.025) and between ATRT-TYR and ATRT-SHH (FDR = 0.016), but not between ATRT-MYC and ATRT-SHH (FDR = 0.706). Expression remains below 11% detection across all subgroups.

However, the absolute magnitude of these differences is small, and expression rates remain below 11% across all subgroups. These results suggest that ZMYM3 is not robustly expressed at the transcript level in any AT/RT subgroup — the statistically significant subgroup differences are unlikely to reflect a biologically or therapeutically meaningful restriction of ZMYM3 activity to the ATRT-TYR lineage. Rather, the modest variation likely reflects lineage-specific differences in CoREST complex utilization that do not rise to the level of subgroup-selective expression.

DISCUSSION

Our integrated analysis identifies ZMYM3 as a high-confidence context-specific fitness gene in AT/RT using a consensus CRISPR dependency screening approach. However, snRNA-seq analysis of primary patient tumors reveals universally low ZMYM3 expression across all cell populations — tumor and non-malignant alike — without meaningful tumor-specific enrichment. This disconnect between functional dependency and expression has important implications for both the mechanistic understanding of ZMYM3 in AT/RT and the broader interpretation of DepMap-identified targets. Of the three consensus hits, we focus on ZMYM3 because SMARCD1's essentiality is biologically expected as a residual SWI/SNF subunit required in SMARCB1-deficient contexts,[10] while NABP2 is a DNA damage repair factor without a clear mechanistic connection to the SWI/SNF–CoREST epigenetic axis that defines AT/RT biology. ZMYM3, by contrast, bridges both complexes, and thus offers a more novel and mechanistically informative hit. Moreover, SMARCD1 exhibits considerably higher dependency scores in non-AT/RT indications, forecasting potential off-target toxicities compared to ZMYM3.

Context-specific fitness genes identified by CRISPR knockout screens reflect functional requirement, not differential expression. ZMYM3, as a component of the CoREST transcriptional repressor complex, may be required at constitutive expression levels to perform a function that is selectively essential in the SMARCB1-deficient context. In normal cells, ZMYM3-mediated CoREST activity operates alongside an intact SWI/SNF complex to maintain transcriptional homeostasis. Following SMARCB1 loss, cBAF-mediated chromatin remodeling is compromised, and residual CoREST repressive machinery — including ZMYM3 — may become disproportionately important for sustaining the aberrant transcriptional programs driving AT/RT tumorigenesis. Under this model, ZMYM3 need not be upregulated; rather, its constitutive expression becomes selectively indispensable because compensatory remodeling can no longer be maintained through alternative mechanisms — consistent with the concept of non-oncogene addiction.

The finding that neurons express higher ZMYM3 than tumor cells (mean = 0.172 vs. 0.102) raises important considerations for therapeutic development. Pharmacological inhibition would need to achieve a therapeutic window based on differential functional dependence rather than differential expression. This parallels EZH2 biology in SMARCB1-deficient tumors: EZH2 is broadly expressed, yet its inhibition selectively impacts SMARCB1-mutant cells uniquely reliant on PRC2-mediated repression.[11] Similarly, ZMYM3 inhibition may selectively disrupt CoREST function in AT/RT cells without equivalent toxicity in non-malignant populations that retain intact SWI/SNF activity. Functional validation will be essential to determine whether this differential dependence translates to a usable therapeutic window.

The modest subgroup-level variation in ZMYM3 expression — with ATRT-TYR showing slightly higher expression than ATRT-MYC and ATRT-SHH — likely reflects lineage-specific differences in CoREST complex utilization rather than a therapeutically meaningful subgroup restriction. Expression remained below 11% of tumor nuclei across all subgroups, and absolute differences in mean expression were small. This sparse pattern is consistent with the known biology of chromatin-associated regulatory factors, which often exert outsized functional effects through stoichiometric roles in multi-protein complexes despite modest transcript levels.

Several limitations warrant acknowledgment. First, snRNA-seq captures nuclear transcriptomes, though for a nuclear chromatin-associated factor like ZMYM3, nuclear RNA abundance is a reasonable proxy for protein expression. Second, transcript-level analysis cannot assess post-transcriptional regulation, protein stability, or complex assembly — any of which could produce differential ZMYM3 activity between tumor and normal cells independent of mRNA differences. Third, the non-malignant reference populations captured within tumor specimens are limited; comparison against matched pediatric or fetal brain atlases (e.g., the Allen Brain Cell Atlas) would provide a more comprehensive off-tumor expression assessment. Fourth, seven patient samples, while sufficient for cell type annotation and expression characterization, limits power to detect subtle subgroup-specific effects.

These findings collectively reframe ZMYM3 as a constitutively expressed gene with context-specific functional essentiality rather than a transcriptionally selective target — a distinction critical for therapeutic strategy. We propose that essential next steps include CRISPR-mediated ZMYM3 knockout in isogenic SMARCB1-proficient and SMARCB1-deficient cell lines to confirm the synthetic-lethal relationship, ChIP-seq to map ZMYM3 binding sites in AT/RT versus normal neural cells, and in vivo assessment in patient-derived xenograft or genetically engineered mouse models to evaluate whether ZMYM3 loss selectively impairs AT/RT growth while sparing normal CNS development. Until such validation is completed, the therapeutic relevance of ZMYM3 as a directly druggable AT/RT target remains an open and actively testable hypothesis.

METHODS

CRISPR DepMap Data Acquisition and Preprocessing. Chronos-corrected CRISPR gene-effect scores and cell line metadata were downloaded from the DepMap 26Q1 public release.[20] The gene-effect matrix comprised 1,208 cell lines and 18,531 genes. Gene names were parsed from the DepMap column format to standard gene symbols. Cell lines were annotated with cancer type using the OncotreeSubtype field from the model metadata. Cell lines annotated as “Atypical Teratoid/Rhabdoid Tumor,” “Rhabdoid Cancer,” or “Extrarenal Rhabdoid Cancer” were merged into a single indication (16 cell lines total) to maximize statistical power. Only indications with at least 4 cell lines were included in the statistical analysis (71 indications).

Context-Specific Gene Identification. For each indication, every gene was tested for context-specific essentiality (stronger fitness defect in the target indication versus all other cell lines) using three complementary statistical tests applied in parallel. (1) Fisher’s exact test: gene-effect scores were binarized as “dependent” (score ≤ -0.6) or “non-dependent,” and a one-sided test assessed enrichment of dependent cell lines in the target indication. (2) Independent two-sample t-test (Welch’s, unequal variance): a one-sided test assessed whether mean scores were more negative in the target indication. (3) Wilcoxon rank-sum test: a one-sided test assessed whether score distributions were stochastically smaller in the target indication; this test was excluded for indications with fewer than 10 cell lines due to insufficient statistical power at small sample sizes. P-values from each test were independently corrected for multiple testing using the Benjamini-Hochberg false discovery rate (FDR) procedure. The top 10 genes passing an FDR threshold of 0.05 were retained per test, and genes appearing in the intersection of all applicable tests were designated consensus context-specific fitness genes.

Bulk Expression Analysis. CCLE RNA-seq expression data was obtained from the same DepMap release. ZMYM3 expression was extracted and merged with cell line cancer type annotations for cross-indication comparison.

Co-Dependency Analysis. Pearson correlation coefficients were computed between the ZMYM3 gene-effect profile and those of all other genes across the AT/RT cell lines. Correlations with members of the CoREST complex (RCOR1, RCOR2, KDM1A, HDAC1, HDAC2, HMG20B, PHF21A, GSE1, ZMYM2, ZMYM4) and SWI/SNF complex (SMARCB1, SMARCA4, SMARCA2, ARID1A, ARID1B, SMARCC1, SMARCC2, SMARCD1, DPF2) were examined. All analyses were performed in Python using pandas, scipy, and statsmodels.

snRNA-seq Data Acquisition. Processed count matrices and cell-level metadata for seven AT/RT patient samples were downloaded from GEO accession GSE283842 (Paassen et al., 2025; 10x Chromium multiome snRNA-seq, Maxima patient cohort).[16] The deposited data had been pre-processed through Cell Ranger ARC and included author-provided quality control annotations, cell type labels, Scrublet doublet scores, UMAP coordinates, and AT/RT subgroup assignments derived from matched bulk DNA methylation profiling.

Quality Control. Although the deposited dataset had undergone prior quality control, we applied confirmatory QC thresholds using Scanpy v1.10.[21] Nuclei were retained if they had 200–6,000 detected genes and fewer than 20% mitochondrial gene fraction — an appropriate threshold for snRNA-seq of frozen tissue, where residual mitochondrial signal primarily reflects damaged nuclei rather than cytoplasmic contamination.[13] Doublets were excluded using the authors' deposited Scrublet scores[22] (threshold = 0.25). After filtering, 9,604 of 9,703 deposited nuclei were retained (98.9%), confirming the quality of the original processing.

Normalization, Feature Selection, and Scaling. Counts were normalized to 10,000 counts per nucleus (CP10K) and log1p transformed. The top 2,000 highly variable genes (HVGs) were identified using Scanpy's mean-dispersion binning method (flavor='seurat').

ZMYM3 was not among the top 2,000 HVGs, consistent with its low and sparse expression; it was nonetheless evaluated across the full normalized expression matrix, as feature plots and per-cluster quantification do not require HVG selection. The count matrix was then scaled to zero mean and unit variance per gene (maximum value clipped at 10) prior to PCA.

Dimensionality Reduction. PCA was performed on the scaled HVG matrix. Thirty PCs were retained based on inspection of the explained variance (elbow) plot.[23]

Batch Correction. Harmony batch correction[18] was applied to the PCA embedding (theta = 2, max iterations = 10), correcting for patient-of-origin as the batch variable. Successful integration was confirmed via post-correction UMAPs: patient-derived nuclei interleaved within biologically coherent clusters, and subgroup separation was maintained in tumor cell clusters.

Clustering. A shared nearest-neighbor (SNN) graph was constructed from the Harmony-corrected PC embedding (n_neighbors = 15, n_pcs = 30). Community detection was performed using the Leiden algorithm (resolution = 0.5).[24] UMAP visualization was computed on the Harmony-corrected embedding (min_dist = 0.3, spread = 1.0).[25]

Cell Type Annotation. Clusters were annotated by computing average expression of canonical CNS and immune marker genes per cluster, running differential expression analysis to identify cluster-specific markers (Wilcoxon test), and cross-referencing against the cell type annotations deposited by Paassen et al. in the sample metadata. Tumor cell clusters (13 clusters, 8,694 nuclei) were distinguished from non-malignant populations (6 clusters, 910 nuclei). Non-malignant cell types included microglia/macrophages (P2RY12, CX3CR1, AIF1), astrocytes (GFAP, AQP4), endothelial cells (PECAM1, VWF), pericytes, and neurons (RBFOX3, SYT1). AT/RT subgroup identity was assigned based on sample-of-origin metadata.

ZMYM3 Expression Analysis. ZMYM3 expression was visualized as a UMAP feature plot (normalized log1p counts) and as violin plots per annotated cell type. Differential expression of ZMYM3 between tumor cells and each non-malignant cell type was assessed using Wilcoxon rank-sum tests (one-sided: tumor > reference) with Benjamini-Hochberg FDR correction. Within the tumor cell fraction, ZMYM3 expression was compared across AT/RT subgroups by Kruskal-Wallis test with post-hoc pairwise Wilcoxon tests and BH-FDR correction.

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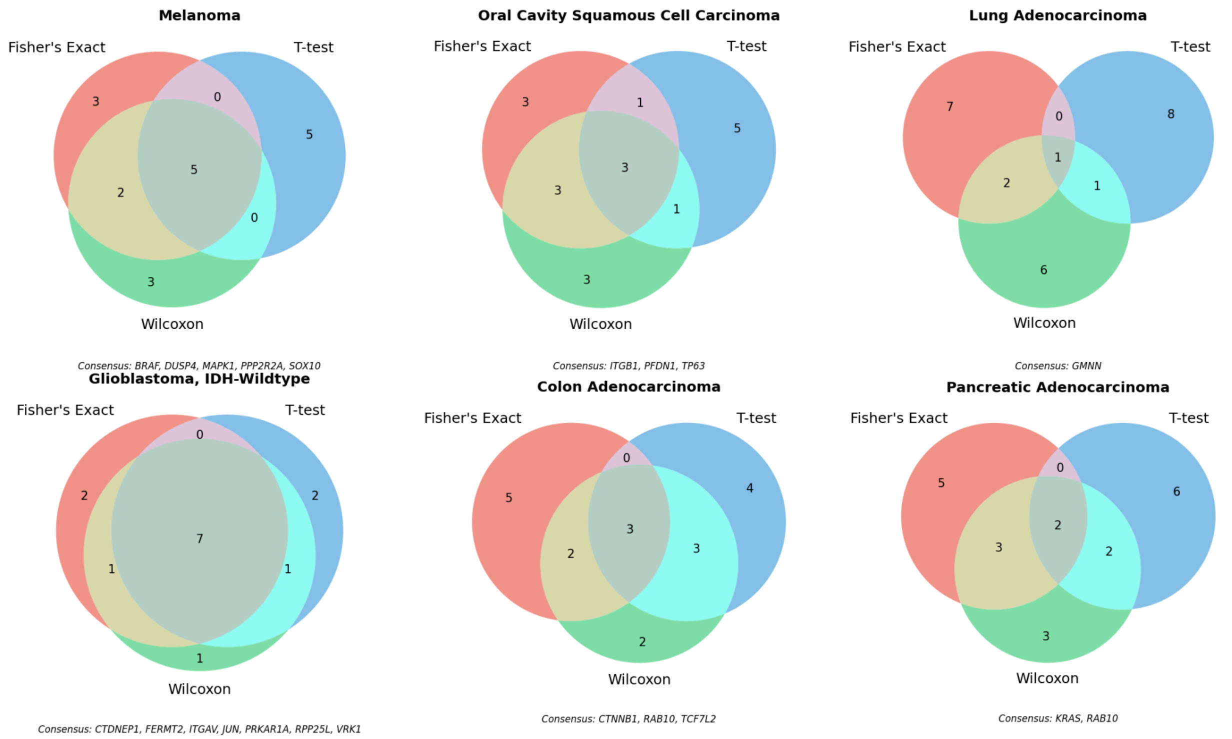

Figure S1. Pan-cancer validation of the consensus dependency framework. Venn diagrams showing the overlap of significant hits from Fisher's exact test, t-test, and Wilcoxon rank-sum test for six representative cancer indications. Consensus hits (triple intersection) recover established lineage dependencies including BRAF/MAPK1/SOX10 in melanoma, CTNNB1 in colon adenocarcinoma, and KRAS in pancreatic adenocarcinoma, validating the approach prior to application to AT/RT.

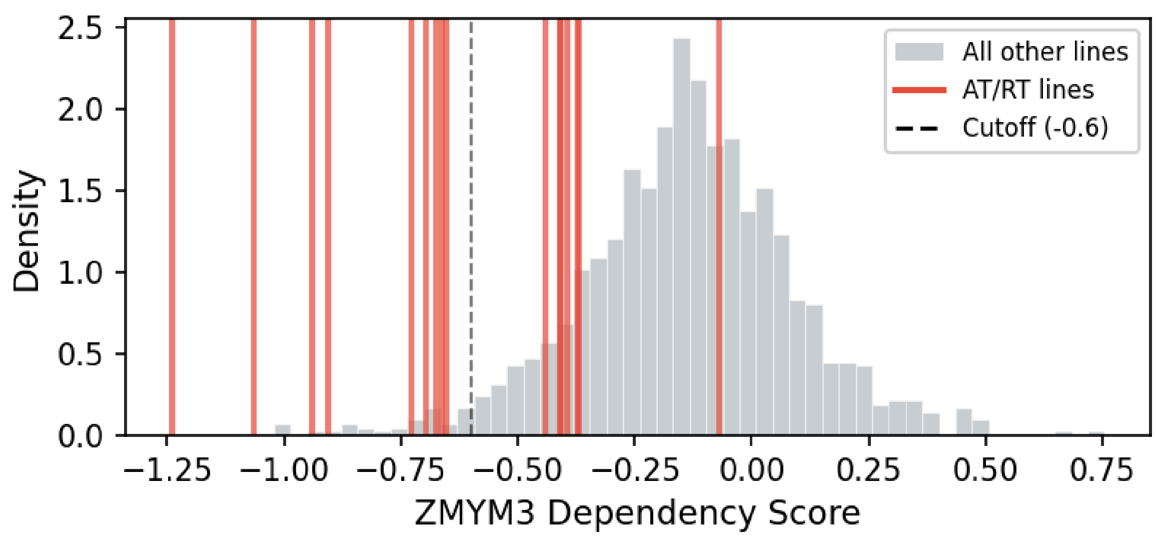

Figure S2. Distribution of ZMYM3 dependency scores across all cell lines in DepMap. Red lines indicate the position of an AT/RT cell line; over half of these cell lines are more negative than the –0.6 cutoff.